



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY
Preliminary Examination
Paper 2

8872/02
19 August 2011
2 hours

Additional Materials: Data Sheet
 Writing Papers

Candidates answer Section A on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.

SECTION A:

Answer **all** questions in the space provided.

SECTION B:

Answer any **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
MCQ P1	
Section A	
B4	
B5	
B6	
Total	

This document consist of **16** printed pages and **2** blank page

Section A

Answer **all** the questions in this section in the spaces provided.

For
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1 Elements **D** and **E** are period 3 elements.

(a) **D** is a white solid with melting point of 317 K. When exposed to air, it reacts spontaneously, producing a white powder of empirical formula D_2O_5 . D_2O_5 readily undergoes neutralisation with aqueous NaOH to form salt and water.

(i) From the observations, deduce an identity of **D**. Include equations, with state symbols, for any reactions that has occurred.

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(ii) Suggest and account for a pH value for the resultant solution when the oxide of **D** dissolves in water.

[6]

- (b) The successive ionisation energies of element **E** is shown below.

	1st	2 nd	3rd	4th	5th	6th	7 th	8 th
I.E. kJmol ⁻¹	999	2265	3331	5071	7008	8487	27107	31719

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- (i) Deduce the group number in which **E** belongs.
- (ii) Write down the electronic configuration of **E** in the +4 oxidation state.

[2]

- (c) Compare the 1st ionisation energy of **E** with the element just before it. Explain.

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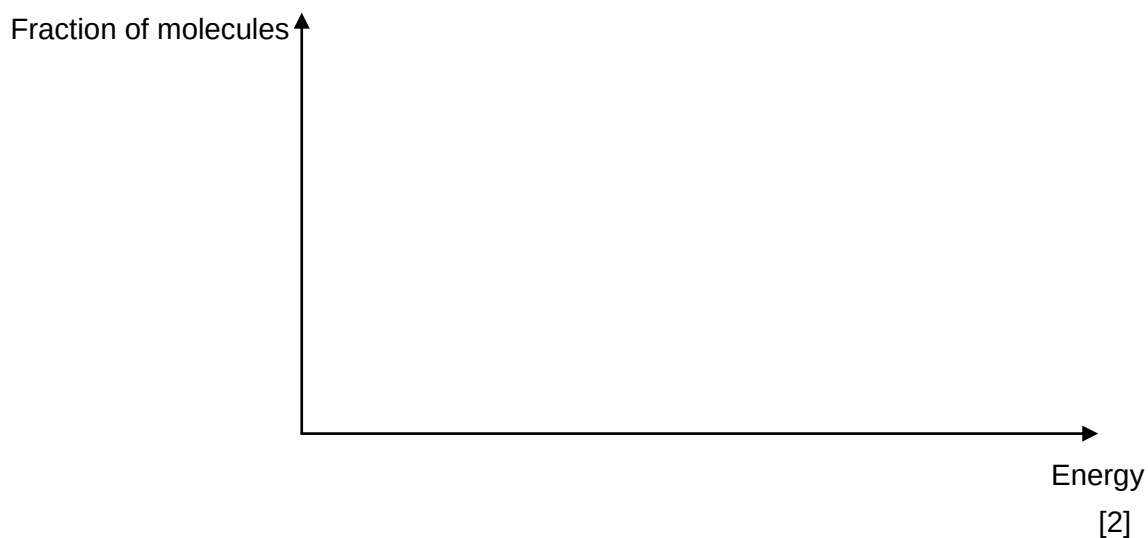
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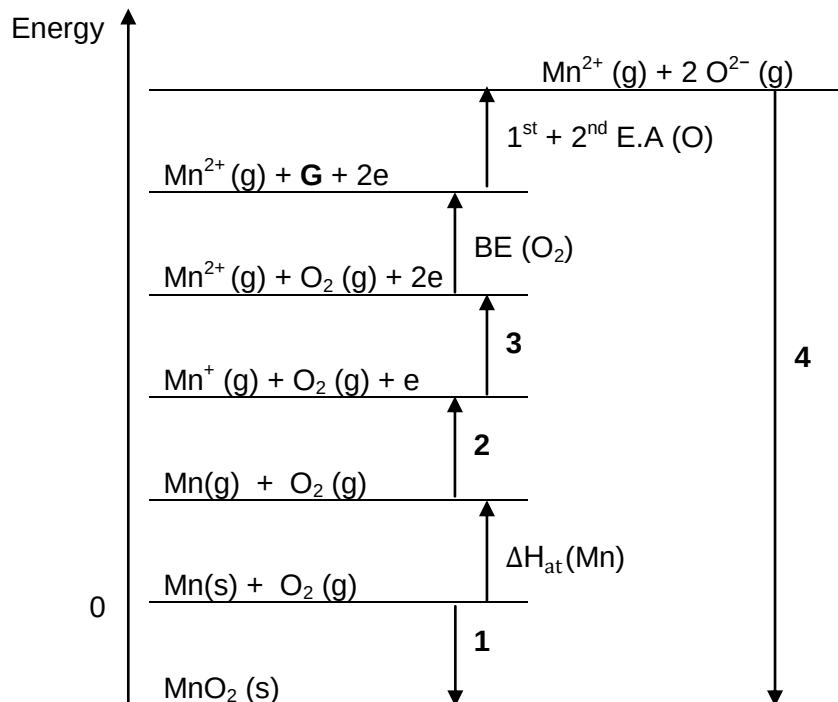
Total: [10]

- 2 Manganese dioxide, MnO_2 is a strong oxidising agent and it is also commonly used as a catalyst.

(a) On the grid below, sketch and label the **energy distribution** of gas molecules in the presence of a catalyst. Your sketch should clearly illustrate the effect of catalyst on rate of reaction.



(b) The lattice energy of MnO_2 can be calculated based on the energy cycle shown below.



(i) Identify the species **G**, with appropriate state symbol

G:

(ii) Identify the enthalpy changes labeled by the numbers **1 to 4**.

1

2

3

4

[3]

(c) Potassium manganate (VII), KMnO_4 , can be manufactured from MnO_2 . During the process, MnO_2 is reacted with concentrated KOH to produce potassium manganate (VI), K_2MnO_4 . When a solution of K_2MnO_4 is acidified, a *disproportionation* reaction occurs. This produces a dark brown MnO_2 precipitate and a solution of KMnO_4 .

(i) With reference to oxidation states of manganese, explain what is meant by *disproportionation* and suggest an equation for this reaction.

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(ii) In alkaline solution, KMnO_4 reacts with sodium sulfite, Na_2SO_3 in a 1:1 ratio. Suggest the final oxidation state of manganese.

[Given: $\text{SO}_3^{2-} + 2\text{OH}^- \rightarrow \text{SO}_4^{2-} + \text{H}_2\text{O} + 2\text{e}^-$]

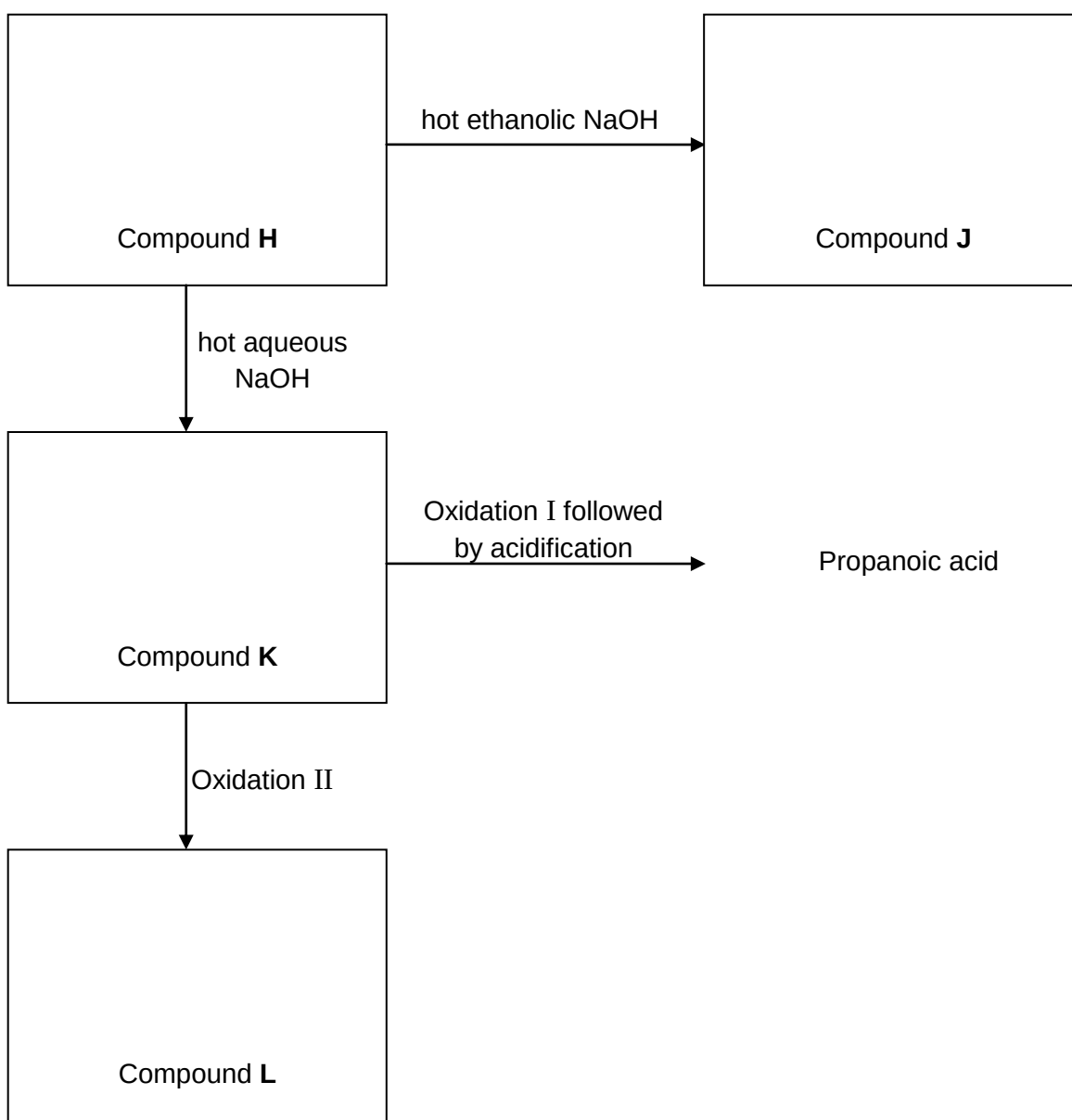
[5]

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- 3** **H** is a halogenoalkane with molecular formula C_4H_9Cl . When reacted with hot ethanolic NaOH, the product, **J** is found to display geometrical isomerism. When refluxed with aqueous NaOH, compound **K** is formed.

K undergoes oxidation when reacted with two different sets of reagents and conditions. When **K** is oxidised using the first set of reagents and conditions, followed by acidification, propanoic acid is formed. However, when **K** is oxidised using the second set of reagents and conditions, another product **L** is formed. **L** reacts with 2,4-DNPH but not with Tollens' reagent.

- (a)** Using the information given above, deduce the structures of compounds **H**, **J**, **K** and **L** and draw the structural formulae of the molecules in the relevant boxes provided in the flowchart.



[4]

- (b)** State the reagents and conditions for Oxidation I and Oxidation II.

Oxidation I:

Oxidation II:

[2]

- (c)** Draw and name the two isomers of compound **J**.

[2]

- (d)** Compound **K** is found to have a higher boiling point as compared to compound **L**.
With reference to structure and bonding, explain why this is so.

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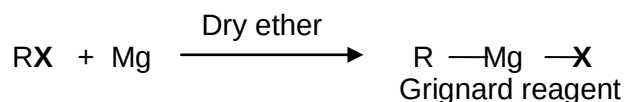
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[2]

Total: [10]

- 4 Halogenoalkanes, RX , are important intermediates used to produce many organic compounds. They may be converted into Grignard reagents by reacting them with magnesium in dry ether:



where R is an alkyl group and X is chlorine, bromine or iodine.

- (a) (i) With relevant data from data booklet, explain why the reactivity between halogenoalkanes and Mg is in the following order:
iodoalkane > bromoalkane > chloroalkane

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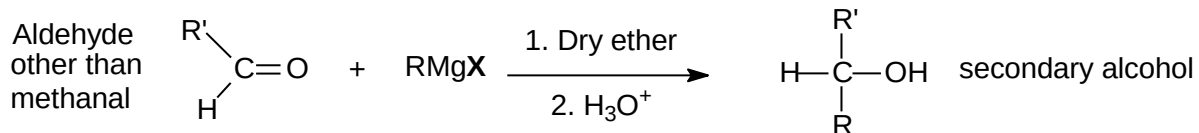
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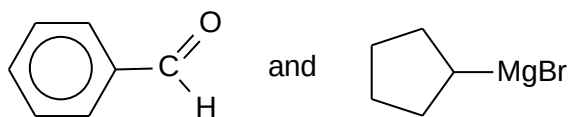
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On reaction with carbonyl compounds, $RMgX$ forms various classes of alcohols. The reaction below shows an example of the formation of a secondary alcohol:



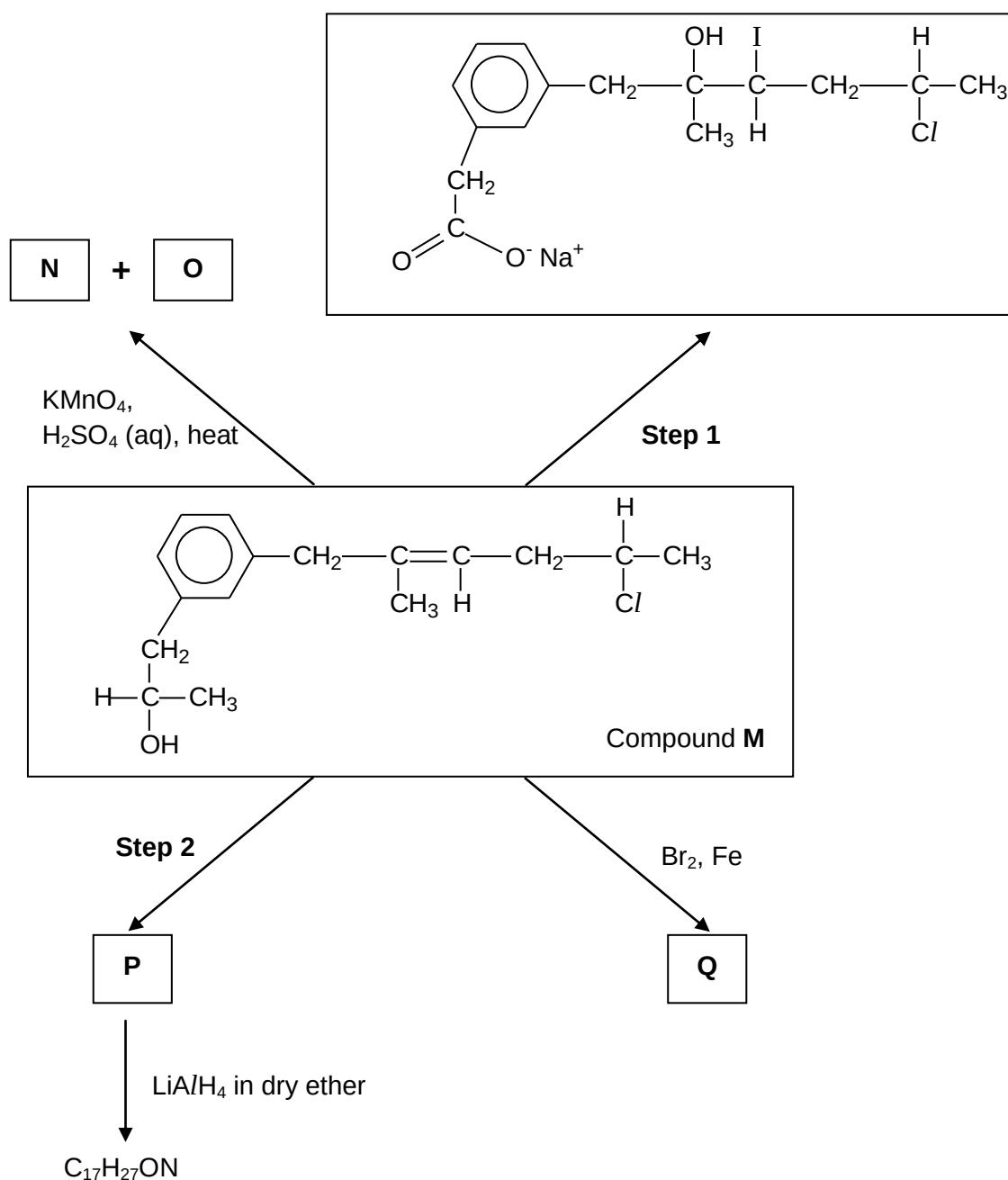
- (ii) Draw the structure of the organic product formed when the following are reacted:



- (iii) Draw the displayed formula of the chloroalkane which can be used to form a Grignard reagent that would react with propanone to give 2-methylpentan-2-ol.

[4]

- (b) The following shows a series of reactions that compound **M** can undergo.



- (i) Draw the structures of compounds **N** to **Q** in the boxes provided.

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- (ii) State the reagents and conditions for **steps 1 and 2**:

Step 1:

Step 2:

[6]

Total: [10]

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Section B

Answer **two** questions from this section on separate answer paper.

- 5 Hydroxy acids are compounds that consist of a carboxylic acid with a hydroxyl group on the adjacent carbon. A typical structure of a hydroxy acid is as follows:



These compounds are most often used in the cosmetics industry to produce anti-wrinkle and anti-ageing products for the skin.

- (a) Compound **S** is a hydroxy acid with the molecular formula $\text{C}_x\text{H}_y\text{O}_3$.
When 7.5 cm^3 of the liquid **S** was reacted with an excess of sodium metal, 2.4 cm^3 of hydrogen gas was produced.
Another sample of 7.5 cm^3 of the liquid **S** was completely combusted in an excess of oxygen in an enclosed vessel, the volume of gas (measured at r.t.p.) did not change. When the mixture of gas was passed through NaOH (aq) , a reduction in gas volume of 7.2 cm^3 (measured at r.t.p.) was observed.

- (i) Write a balanced equation for the combustion of **S**.
(ii) Use the data given above to calculate values for x and y in the molecular formula of compound **S**.

[5]

- (b) Compound **T** is another hydroxy acid with molecular formula $\text{C}_8\text{H}_8\text{O}_3$. It reacts with hot, acidified KMnO_4 to produce **U** and carbon dioxide gas.

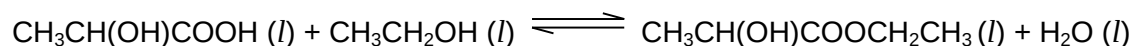
When **T** is refluxed with a small amount of concentrated sulfuric acid, a molecule, **V** is produced.

V has 3 six-membered rings and does not react with sodium metal or sodium carbonate. **V** has a molecular formula of $\text{C}_{16}\text{H}_{12}\text{O}_4$.

Deduce the structural formulae of **T**, **U** and **V**, giving reasons for your answer.

[6]

- (c) Hydroxy acids can undergo nucleophilic substitution reaction with another organic molecule to produce an ester. An example of such a reaction is shown below:

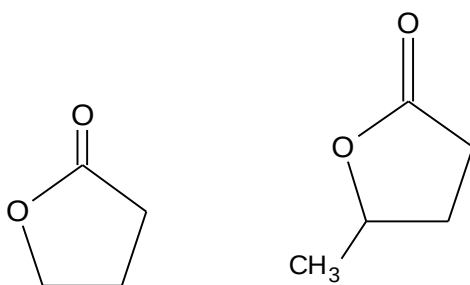


When 23.0 g of $\text{CH}_3\text{CH}_2\text{OH}$ was heated with 40.0 g of $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ in a vessel of $V \text{ dm}^3$ and allowed to reach equilibrium again, it was found that 25% of $\text{CH}_3\text{CH}_2\text{OH}$ remained in the system.

- (i) Define the term *dynamic equilibrium*.
- (ii) Given that the temperature of the system remained unchanged throughout the whole process, calculate the value of K_c .
- (iii) Given that the enthalpy change of the above process is positive, suggest with a reason how the position of equilibrium might alter when the temperature is increased.

[6]

- (d) Apart from undergoing esterification with another organic molecule, hydroxy acids can form cyclic esters (lactones) through a process known as lactonisation. In a lactonisation process, there was some contamination resulting in the production of a mixture of two lactones:

Lactone **W**Lactone **Y**

Describe chemical test(s) by which the two lactones, **W** and **Y** can be distinguished from each other, and describe clearly what will be observed for each of the lactones.

[3]

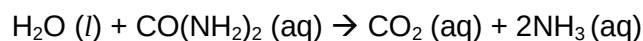
[Total: 20]

- 6 Ammonia, NH_3 , is one of the most highly produced inorganic chemicals. It can be used in fertilisers, cleaning products, colour dyes and even in the making of explosives.

- (a) Ammonia is synthesised from its elements in the Haber process.
- (i) Given that the reaction is exothermic and incomplete, write an equation for the Haber process.
- (ii) State the **three** conditions necessary for the efficient working of a Haber process
- (iii) Explain why the temperature you have quoted in (a)(ii) is used.

[5]

- (b) Ammonia is given off from urine. And the nitrogen containing substance in urine is urea, $\text{CO}(\text{NH}_2)_2$.



Compound	$\Delta H_f^\theta / \text{kJ mol}^{-1}$
$\text{H}_2\text{O} (l)$	- 287.0
$\text{CO}(\text{NH}_2)_2 (\text{aq})$	- 320.5
$\text{CO}_2 (\text{aq})$	- 414.5
$\text{NH}_3 (\text{aq})$	- 81.0

Use the data given below to calculate the standard enthalpy change for the hydrolysis of urea.

[2]

- (c) The labels of two monoacidic bases, aqueous sodium hydroxide, NaOH , and aqueous ammonia, NH_3 , have fallen off in the laboratory.

In order to determine the identity of these chemicals, a student conducted calorimetric experiments to determine the standard enthalpy change of neutralisation with hydrochloric acid with each of these chemicals.

- (i) Define what is meant by the *standard enthalpy change of neutralisation*.
- (ii) Explain why the student is able to distinguish $\text{NaOH} (\text{aq})$ from $\text{NH}_3 (\text{aq})$ from the calorimetric experiments? Assume that both experiments were conducted under similar experimental conditions.
- (iii) When 40 cm^3 of 3.0 mol dm^{-3} hydrochloric acid, was added to 60 cm^3 of 1.4 mol dm^{-3} of one of the monoacidic base in a polystyrene cup. The maximum temperature rise was recorded as 10.5°C .
Given that the specific heat capacity of the solution = $4.2 \text{ J g}^{-1} \text{ K}^{-1}$ and the process is 100% efficient, calculate the standard enthalpy change of neutralisation for the reaction.

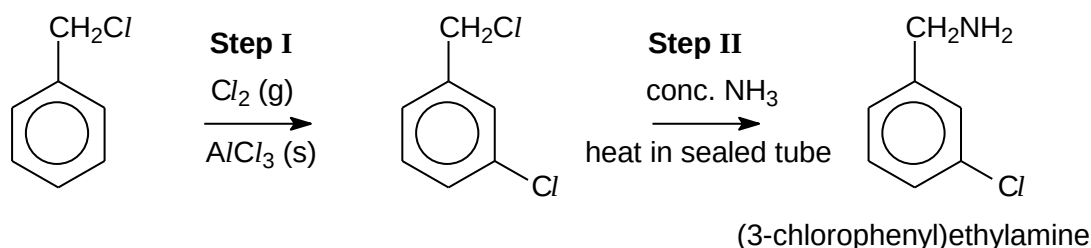
[6]

(d) Phosphine, PH_3 has a similar structure to NH_3 .

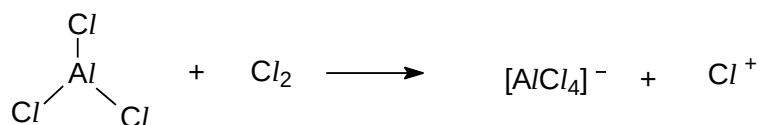
- (i) By considering the VSEPR theory, comment on the relative bond angle of PH_3 and NH_3 .
- (ii) PH_3 has a boiling point of -87.7°C while NH_3 has a boiling point of -33.34°C . Explain for the observation in terms of bonding.

[4]

(e) In organic chemistry, ammonia can act as a nucleophile in substitution reactions to form amines. The following is a proposed synthesis of an amine, (3-chlorophenyl)ethylamine.



In **step I**, AlCl_3 is involved in the following manner:

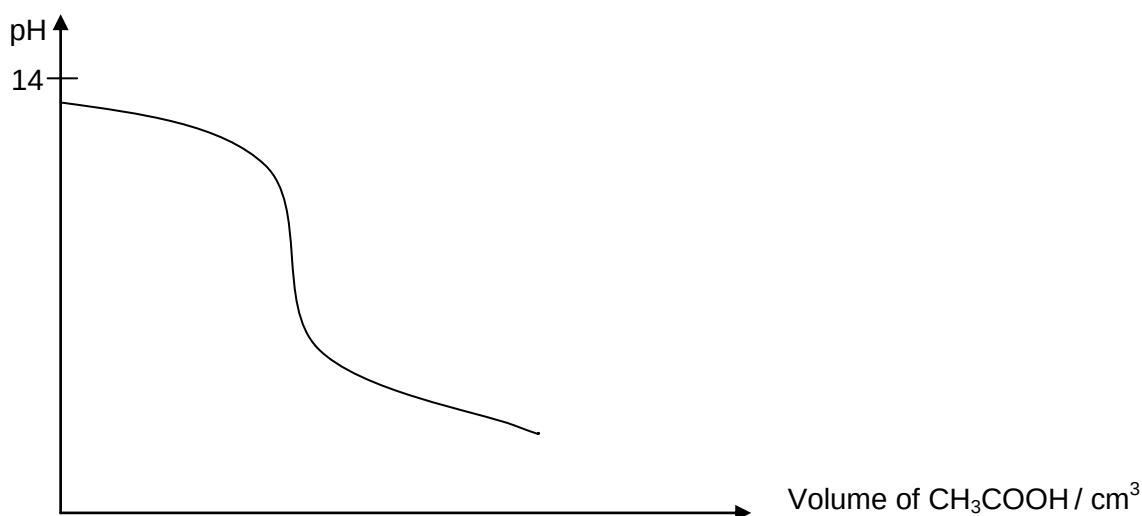


- (i) Draw the dot and cross diagram for $[\text{AlCl}_4]^-$.
- (ii) Using your knowledge in periodicity and with the aid of equations, suggest why AlCl_3 cannot be in aqueous state for it to function in **step I**.
- (iii) Suggest why (3-chlorophenyl)ethylamine cannot be obtained using the above synthesis.

[3]

Total: [20]

- 7 (a) The graph below shows the change in pH when 0.10 mol dm^{-3} of ethanoic acid is gradually added to 10 cm^3 of 0.20 mol dm^{-3} potassium hydroxide.



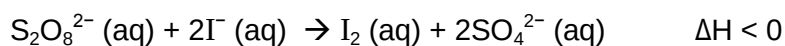
- (i) Calculate the initial pH of the potassium hydroxide solution.
- (ii) Calculate the equivalence volume of ethanoic acid required in this neutralisation.
- (iii) Given the pH range of some acid-base indicators in the table below, select the most suitable indicator for this titration and explain your choice.

Indicator	pH range at which colour change occurs
Methyl violet	0.5 – 1.5
Bromocresol blue	3.8 – 5.5
Alizarin yellow	10.1 – 12.0

- (iv) An equal volume of CH_3COOH and $\text{CH}_3\text{COO}^-\text{Na}^+$ was mixed together. Explain, with the aid of equations, how this mixture can resist changes in pH when small amounts of acid or base is added to it.
- (v) Suggest, with reasoning, whether you would expect the same volume of carbon dioxide gas to be evolved in each case when equal volumes of the CH_3COOH and hydrochloric acid of the same concentration are separately reacted with an excess of sodium carbonate.
- (vi) Compare the acidities of ethanol, ethanoic acid and chloroethanoic acid. Explain your answer.

[14]

(b) Peroxodisulfate ions, $\text{S}_2\text{O}_8^{2-}$, can oxidise iodide ions, I^- as shown in the equation:



The kinetics of this reaction can be investigated experimentally by varying the concentration of the two reactants involved and determining the time taken for the brown iodine colour to first appear. The data below was obtained.

Expt	$[\text{S}_2\text{O}_8^{2-}] / \text{mol dm}^{-3}$	$[\text{I}^-] / \text{mol dm}^{-3}$	Relative rate / s^{-1}
1	0.1	0.2	$\frac{1}{32}$
2	0.2	0.2	$\frac{1}{16}$
3	0.3	0.3	$\frac{9}{64}$

- (i) Deduce the order of reaction with respect to $\text{S}_2\text{O}_8^{2-}$ and I^- . Explain your reasoning.
- (ii) Write the rate equation for this reaction.
- (iii) Suggest a reason why the reaction is slow.
- (iv) Define rate constant, k , and state the effect on k when temperature is lowered.

[6]

[Total: 20]

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